Graph-Guided Domain-Adversarial Learning for Neonatal Sepsis Biomarker Discovery

Student Researcher | ACSEF 2026

Objective

Can a biology-constrained multiplex hypergraph model improve external diagnosis of neonatal sepsis while preserving biomarker interpretability?

Methods

Data: GSE25504 + GSE69686 for development, GSE26440 for external validation.
Pipeline: harmonization, batch correction, multiplex hypergraph relations (KEGG/STRING/Co-expression), relation attention, gene-scoring mask, MLP classifier, domain-adversarial head.

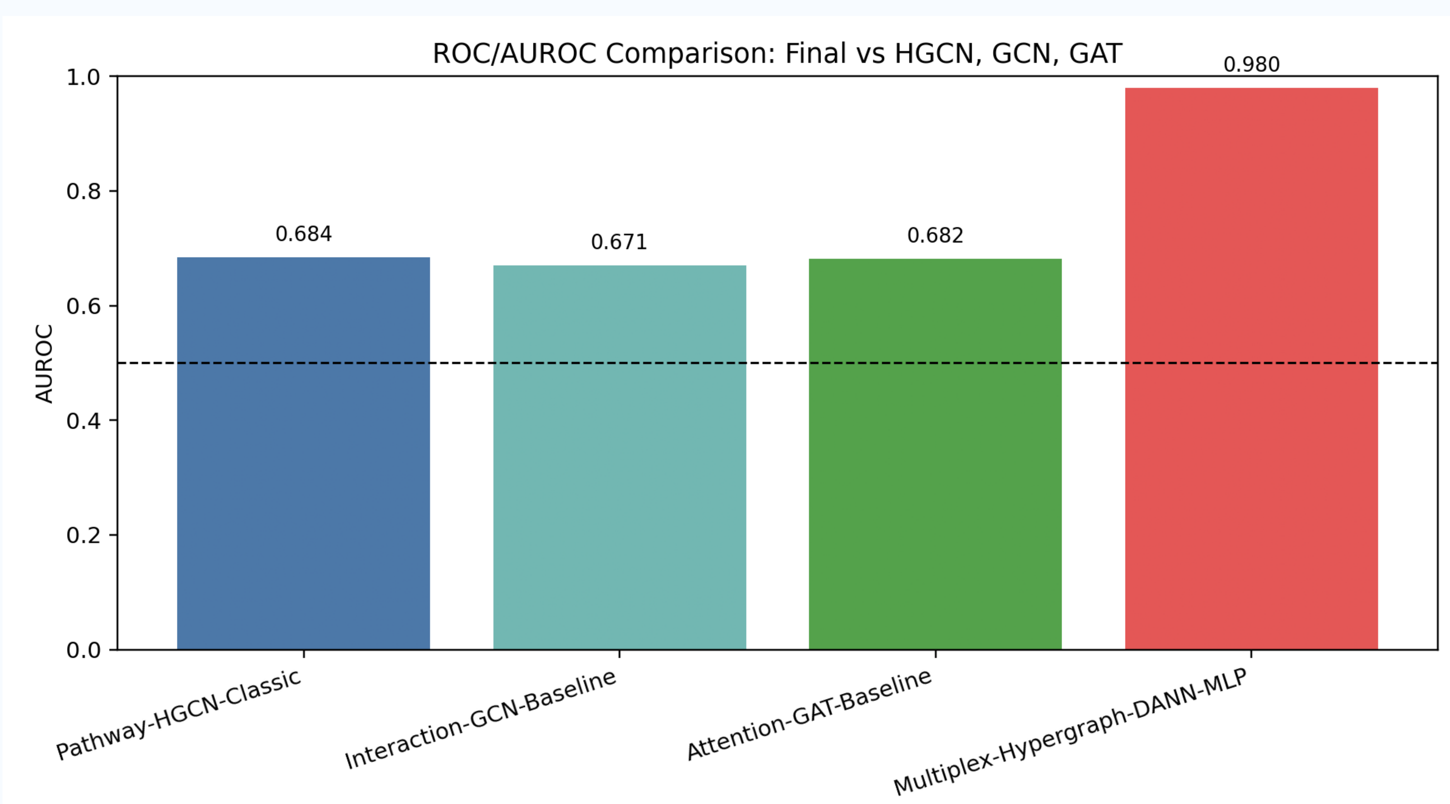
Primary Results

5-fold sepsis CV: AUROC 0.9796, Accuracy 0.9779, F1 0.9733. External GSE26440: AUROC 0.9856, Accuracy 0.9519, F1 0.9697. Top biomarkers include TNFAIP6, S100A12, RETN, and CD52.

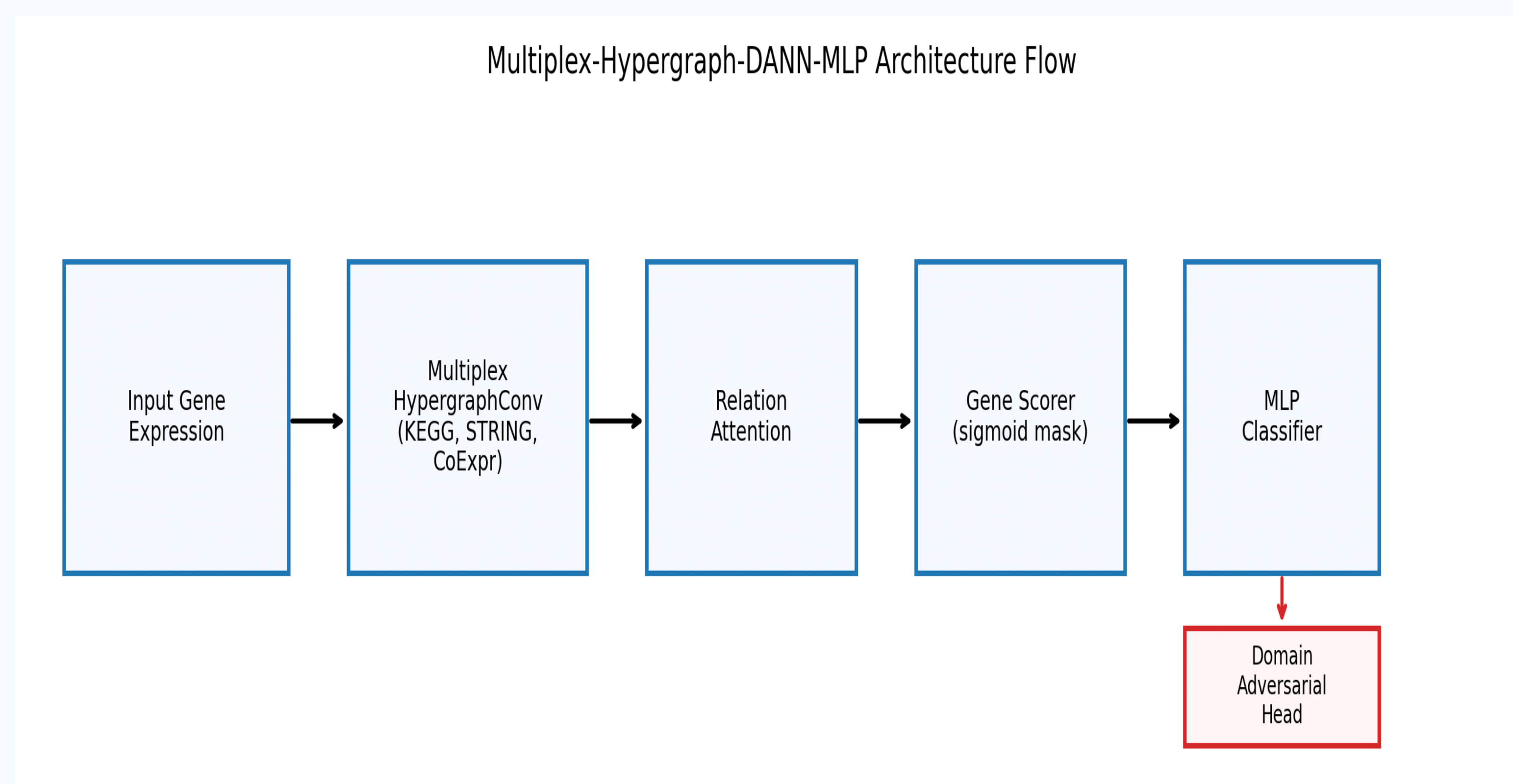
Conclusions

Multiplex-Hypergraph-DANN-MLP outperformed HGCN/GCN/GAT baselines and remained stable externally. Transfer experiments in osteogenesis imperfecta support broader rare-disease applicability.

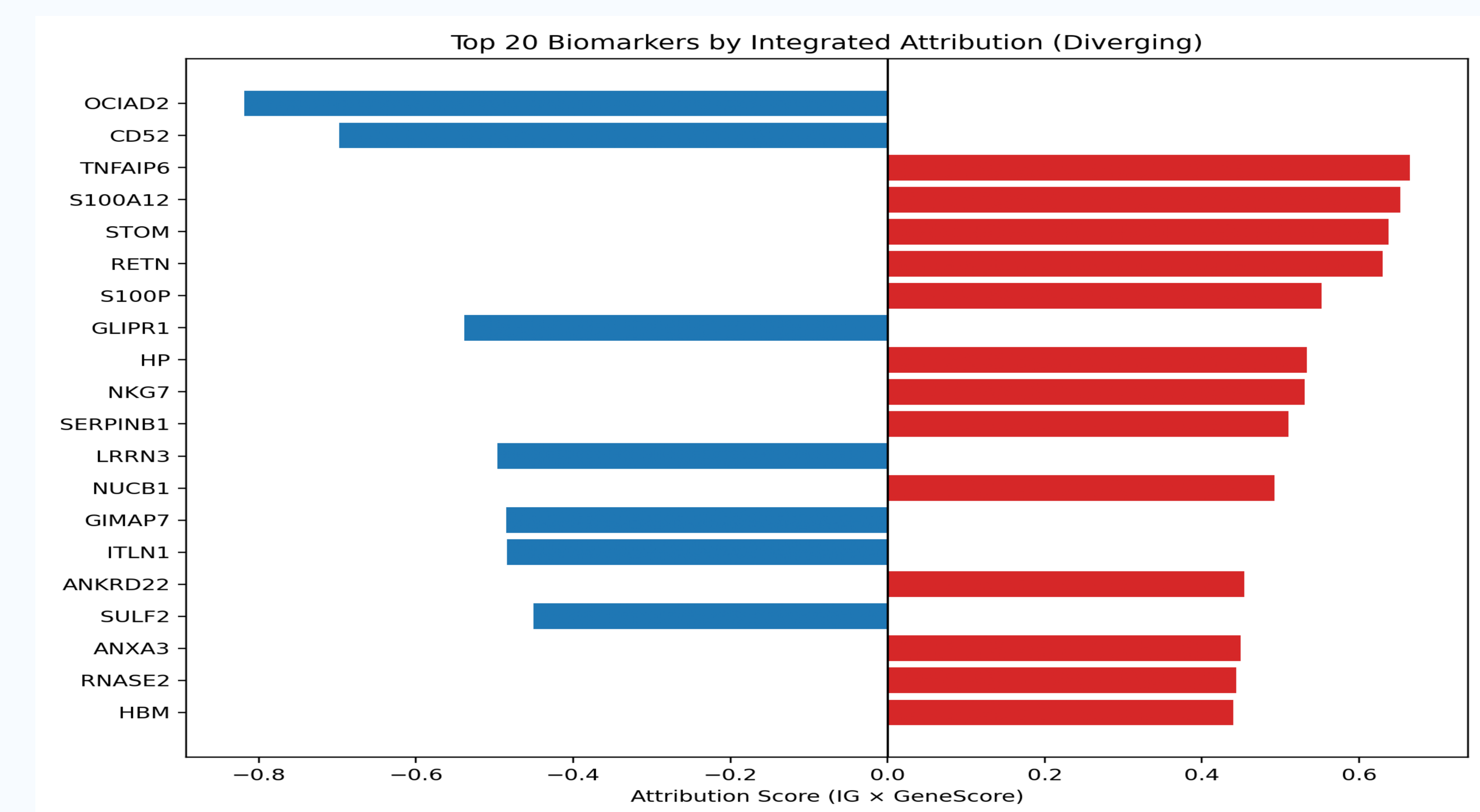
A. ROC Comparisons



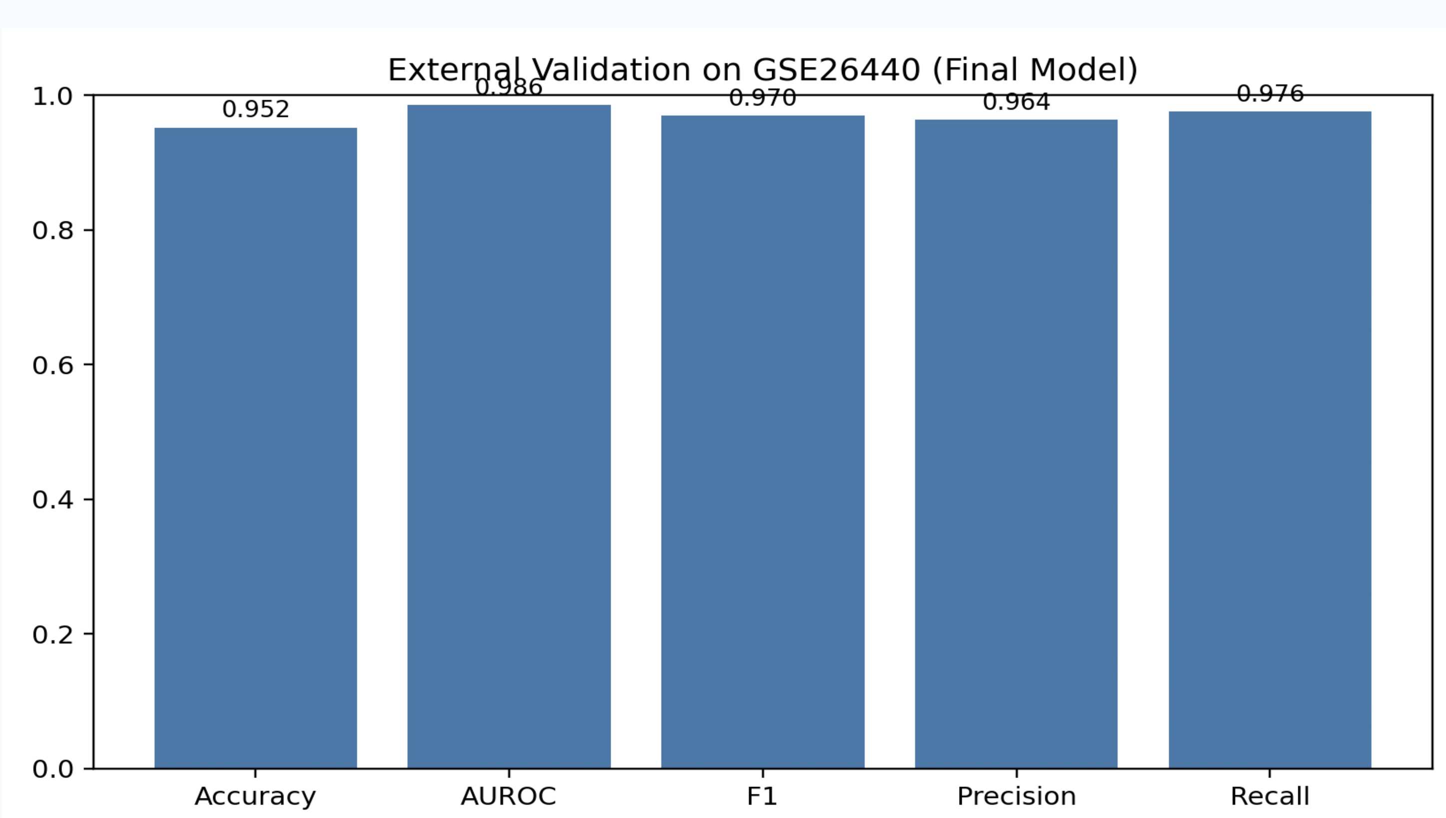
B. Architecture Flowchart



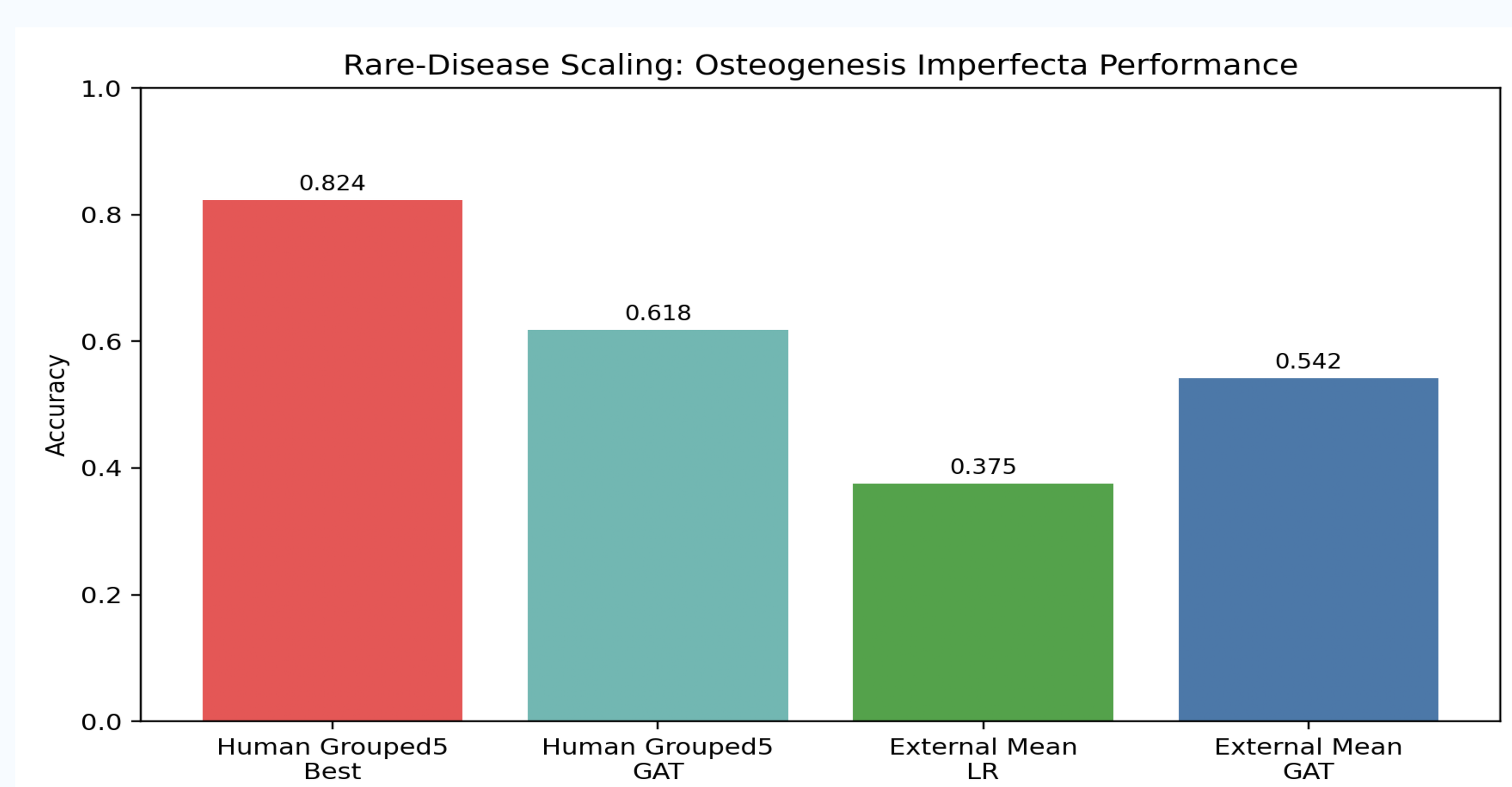
C. Biomarker Attributions



D. External Validation



E. OI Scaling Summary



F. Model Metric Radar